

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Reads graph and uses $10^{\log P}$ or $10^{\log V}$ to get either P or V	AO3.4	M1	$\log_{10} P = 2.18$ $P = 151$ $\log_{10} V = -0.15$ $V = 0.708$
	Correctly obtains P and V AWRT 150 and AWRT 0.71	AO1.1b	A1	
(b)	Calculates value of gradient to find d Condone use of $\log d =$ gradient Or uses a value of c plus a P/V pair to find d	AO3.4	M1	$\log_{10} P = \log_{10} c + d \log_{10} V$ Gradient = $d = -1.4$ Intercept = $\log_{10} c$ $c = 93.3$
	Obtains correct value for d AWRT -1.4 Not necessarily a decimal	AO1.1b	A1	
	Calculates value of intercept to find $\log_{10} c$ Or uses a value of d plus a P/V pair to find c	AO3.4	M1	
	Calculates correct value for c AWRT 93	AO1.1b	A1	
(c)	Uses their values of c and d in the formula $P = cV^d$	AO1.1a	M1	$P = 93.3 \times 2^{-1.4}$ $= 35.4$ kilopascals
	Obtains P value, including units AWFW 30 to 40	AO3.2a	A1	
				Total 8 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Expresses $2\log_a 6$ as $\log_a 6^2$ or $\log_a 36$ divides by 3 inside the log term (PI)	AO1.1a	M1	$\log_a \left(\frac{6^2}{3} \right)$

e.g. $2\log_a\left(\frac{6}{3}\right)$ scores M1 but not $2\frac{\log_a 6}{\log_a 3}$			$\log_a 12$
Obtains correct final answer (condone the omission of base a) CAO	AO1.1b	A1	
Total 2 marks			

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses $n \log_a x = \log_a x^n$ correctly	AO1.1a	M1	$\log_a y = 2\log_a 7 + \log_a 4 + \frac{1}{2}$
	Uses $\log_a x + \log_a y = \log_a xy$ or $\log_a x - \log_a y = \log_a \frac{x}{y}$ correctly	AO1.1a	M1	$\Rightarrow \log_a y = \log_a 7^2 = \log_a 4 + \frac{1}{2}$ $= \log_a (49 \times 4) + \frac{1}{2}$ $= \log_a 196 + \frac{1}{2} \log_a a$
	Obtains $\sqrt{a}$	AO1.1b	B1	$= \log_a 196 + \log_a \sqrt{a}$
	Obtains correct answer in any correct form.	AO1.1b	A1	$= \log_a 196\sqrt{a}$ $\therefore y = 196\sqrt{a}$
(b)	Explains that $-\frac{3}{2}$ should be rejected as it is not possible to evaluate $\log_a\left(-\frac{3}{2}\right)$	AO2.3	E1	$-\frac{3}{2}$ should be rejected as it is not possible to evaluate $\log_a\left(-\frac{3}{2}\right)$
Total 5 marks				

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains (at least four) correct $\log_{10} y$ values, in table or plotted	AO1.1a	M1	(1, 1.1) (2, 1.7) (3, 2.1) (4, 3.0) (5, 3.1) (6, 3.5)
	Plots all points correctly	AO1.1b	A1	(Points above plotted on

			grid)	
(b)	Identifies $y = 1100$ and gives correct reason	AO2.2b	B1	(4, 1100), as it is not on the line that the other points are close to
(c)	Uses laws of logs. (May earn in part (a) if laws of logs were used there)	AO1.1a	M1	$\log_{10} y = \log_{10} k + x \log_{10} b$ Vertical intercept $c = 0.68$ ($= \log_{10} k$) Therefore from intercept: $k = 10^{0.68}$
	Draws straight line and calculates/measures the vertical intercept c and attempts 10^c or calculates/measures gradient m and attempts 10^m Alternatively uses regression line from calculator to get intercept and gradient	AO1.1a	M1	Gradient $m = 0.48 = \log_{10} b$ Therefore from gradient: $b = 10^{0.48}$
	Finds correct value of b from 'their' gradient, provided $0.45 < \text{'their' gradient} < 0.51$	AO1.1b	A1F	$k = 4.8$
	Finds correct value of k from 'their' intercept, provided $0.6 \leq \text{'their' intercept} \leq 0.8$	AO1.1b	A1F	$b = 3.0$
Total 7 marks				

Q5.

Marking Instructions	AO	Marks	Typical Solution
Correctly applies a single law of logs with either term	AO1.1a	M1	$\log_a(a^3) + \log_a\left(\frac{1}{a}\right) = 3 + (-1)$ $= 3 - 1$ $= 2$
States correct final answer (NMS scores full marks)	AO1.1b	A1	

Total 2 marks

Q6.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Translates proportionality into a differential equation involving $\frac{dA}{dt}$, A and a constant of proportionality.	AO3.3	M1	$\frac{dA}{dt} \propto A$ $\Rightarrow \frac{dA}{dt} = kA$ $\Rightarrow \int \frac{1}{A} dA = \int k dt$
	Separates variables	AO1.1a	M1	$\Rightarrow \ln A = kt + c$ $\Rightarrow A = e^{kt+c}$
	Integrates both of 'their' sides correctly	AO1.1b	A1F	$\Rightarrow A = Be^{kt}$ AG
	Constructs a rigorous mathematical argument that supports use of the given model. AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.	AO2.1	R1	
(b)	States correct value of B	AO1.1b	B1	$B = 0.25$ or $B = \frac{1}{4}$
(ii)	Uses $t = 20$ and $A = 0.5$ to find k	AO3.1b	M1	When $t = 20$, $A = 0.5$ $\Rightarrow 0.5 = 0.25e^{20k}$
	Finds correct value of k	AO1.1b	A1	$\Rightarrow 20k = \ln 2$ $\Rightarrow k = \frac{1}{20} \ln 2$
	Substitutes 'their' k to get A in terms of t	AO1.1a	M1	$\Rightarrow A = \frac{1}{4} (e^{\ln 2})^{\frac{t}{20}}$
	Constructs rigorous and convincing argument to show $A = 2^{\frac{t}{20}-2}$ Using correct notation throughout. AG	AO2.1	R1	$\Rightarrow A = 2^{-2} \times 2^{\frac{t}{20}}$ $A = 2^{\frac{t}{20}-2}$ AG
(iii)	Uses the model to set up correct equation and	AO3.4	M1	$2\pi = 2^{\frac{t}{20}-2}$

attempt to find t			$t = 93.03$ days
Finds correct value of t	AO1.1b	A1	

(c)	States any sensible and relevant limitation of the model that is specified in terms of the pond, area, weed, rate of change or time.	AO3.5b	E1	Model predicts that the area of weed will increase without limit and this is not possible since the area of the pond is 4π
(d)	Any sensible and relevant refinement to the model that is specified in terms of the pond, area, weed, rate of change or time	AO3.5c	E1	Introduce a limiting factor such as fish eating weed or rate of growth decreases as surface area covered
Total 13 marks				

Q7.

$$\log_a N - \log_a x = \frac{3}{2}$$

$$\log_a \frac{N}{x} = \frac{3}{2}$$

A log law used correctly. PI by next line.

M1

$$\frac{N}{x} = a^{\frac{3}{2}}$$

Logarithm(s) eliminated correctly

m1

$$x = a^{-\frac{3}{2}} N$$

ACF of RHS

A1

[3]

Q8.

(a) $b = a^c$

B1

1

(b) $2 \log_2 (x + 7) - \log_2 (x + 5) = 3$
 $\log_2 (x + 7)^2 - \log_2 (x + 5) = 3$

A law of logs used correctly on a correct expression.

M1

$$\log_2 \frac{(x+7)^2}{x+5} = 3$$

A further correct use of law of logs on a correct expression.

M1

$$= 3 \log_2 2 = \log_2 2^3$$

$$\Rightarrow \frac{(x+7)^2}{x+5} = 2^3$$

$3 = 3 \log_2 2$ or $3 = \log_2 2^3 (= \log_2 8)$ seen
 or

eg $\log f(x) = 3 \Rightarrow f(x) = 2^3 (= 8)$ OE

B1

$$\Rightarrow (x + 7)^2 = 8(x + 5)$$

Correct equation having eliminated logs and fractions

A1

$$\Rightarrow x^2 + 14x + 49 = 8x + 40$$

$$\Rightarrow x^2 + 6x + 9 (= 0)$$

A1

Since $6^2 - 4(1)(9) = 0$, (there is only) one value of x
 (which satisfies the given equation).

OE

CSO Need conclusion which is also correctly justified

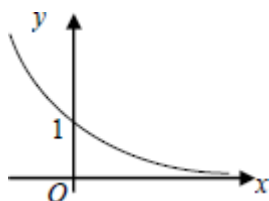
A1

6

[7]

Q9.

(a)



*Correct shaped graph in 1st two quadrants only and
 indication of correct behaviour of curve for large positive*

and negative vals. of x . Ignore any scaling on axes.

B1

y -intercept indicated as 1 on diagram or stated as intercept = 1 or as coords (0, 1).

B1

2

(b) $\frac{1}{2^x} = \frac{5}{4} \Rightarrow 2^{-x} = \frac{5}{4}$ (or $2^x = \frac{4}{5}$ or $2^{2-x} = 5$)

Correct 'rearrangement' to eg $2^x = \frac{4}{5}$ or $2^{-x} = \frac{5}{4}$ or
 $0.5^x = 1.25$ PI or $\log 1 - \log 2^x = \log(5/4)$ or better

M1

$$\log 2^{-x} = \log 1.25 \Rightarrow -x \log 2 = \log 1.25$$

$$[\log 2^x = \log 0.8 \Rightarrow x \log 2 = \log 0.8]$$

$$[\log 2^{2-x} = \log 5 \Rightarrow (2-x) \log 2 = \log 5]$$

$$[2^x = 0.8, x \log_2 0.8]; [0.5^x = 1.25, x = \log_{0.5} 1.25]$$

Takes logs of both sides of eqn of form either $2^x = k$ or
 $2^{-x} = k$ OE and uses 3rd law of logs or log to base 2
 (or base $\frac{1}{2}$) correctly

M1

$$x = -0.321928 \dots \text{ so } x = -0.322 \text{ (to 3sf)}$$

Condone > 3sf [Logs must be seen to be used otherwise
max of M1M0A0]

A1

3

(c) $\log_a b^2 + 3 \log_a y = 3 + 2 \log_a \left(\frac{y}{a} \right)$

$$\log_a b^2 + 3 \log_a y = 3 + 2 [\log_a y - \log_a a]$$

A log law used correctly; condone missing base a .

M1

$$\log_a b^2 + 3 \log_a y = 3 - 2 \log_a a$$

A different log law used correctly condone missing base a .

M1

$$\log_a b^2 y = 3 - 2(1) \quad [\text{or } \log_a b^2 y + \log_a a^2 = 3]$$

Either a further different log law used correctly condone missing base a or $\log_a a = 1$ a stated / used.

M1

$$\Rightarrow \log_a b^2 y = 1 \quad \Rightarrow b^2 y = a$$

$\log Z = k \Rightarrow Z = a^k$ used or a correct method to eliminate logs (dep on no misapplication of any log law OE in the whole solution) Rearrangements which require only two of the above Ms to eliminate logs correctly: award the remaining M with the m mark.

M1

$$\Rightarrow y = ab^{-2}$$

ACF of RHS

A1

5

[10]

Q10.

(a) $\log_k x^2 - \log_k 5 = 1$

A valid law of logs used correctly

M1

$$\log_k \frac{x^2}{5} = 1$$

*Another valid law of logs used correctly
or correct method to reach $\log f(x) = \log 5k$*

M1

$$\log_k \frac{x^2}{5} = \log_k k \quad [\text{or } \log x^2 = \log 5k]$$

PI by next line

A1

$$\Rightarrow \frac{x^2}{5} = k \text{ ie } k = \frac{x^2}{5}$$

Accept either of these two forms.

A1

4

(b) $\log_a y = \frac{3}{2} \quad \log_4 a = b + 2$

$\Rightarrow y = a^{\frac{3}{2}} \quad \Rightarrow a = 4^{b+2}$

For either equation

M1

$y = (4^{b+2})^{\frac{3}{2}}$

*Elimination of a from two correct equations
not involving logarithms*

m1

$y = 2^{3(b+2)}; \quad y = 2^{3b+6}$

CSO Either form acceptable

A1

3

[7]

Q11.

(a) $\{S_{\infty} = \} \quad \frac{a}{1-r} = \frac{12}{1-\frac{3}{8}}$

$\frac{a}{1-r}$ used

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M1

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$\{S_{\infty} = \} 19.2$

19.2 OE NMS mark as 2/2 or 0/2

A1

2

(b) $\{6\text{th term} = \} ar^{6-1}$

Stated or used

M1

$= 12 \times \left(\frac{3}{8}\right)^5 = 2 \times 2 \times 3 \times \left(\frac{3}{2 \times 2 \times 2}\right)^5$

*Changing 8 and 12 in correct expression to
correct products/powers of 2 and 3*

m1

$$= \frac{2 \times 2 \times 3 \times 3^5}{(2^3)^6} = \frac{2^2 \times 3^6}{2^{18}} = \frac{3^6}{2^{16}}$$

AG Be convinced

A1

3

(c) (i) $\{u_n\} = 12x\left(\frac{3}{8}\right)^{n-1}$
 OE. eg $32(3/8)^n$

B1

1

(ii) $\log u_n = \log 12 + \log\left(\frac{3}{8}\right)^{n-1}$

$$\log u_n = \log 12 + (n-1)\log\left(\frac{3}{8}\right)$$

$$\log u_n = \log 12 + (n-1)[\log 3 - \log 8]$$

Log laws

$$\log(PQ) = \log P + \log Q ;$$

$$\log\left(\frac{P}{Q}\right) = \log P - \log Q$$

$$\log(P)^k = k \log P$$

Using (c)(i) and taking logs: one log law used correctly, on a correct expression for u_n

M1

a second different log law used correctly,
 indep of prev M error and ft on cand's (c)(i)
 provided cand's u_n expression has a power
 involving n .

M1

$$\log u_n = \log 3 + 2 \log 2 + (n-1)[\log 3 - 3 \log 2]$$

A third different log law used correctly (or
 equivalent valid step) to reach a correct RHS
 whose terms are all multiples of $\log 2$ and $\log 3$.
 Dep on both prev two Ms

m1

$$\log u_n = n \log 3 - 3n \log 2 + 5 \log 2$$

$$\log_a u_n = n \log_a 3 - (3n - 5) \log_a 2$$

CSO AG Be convinced, no slips although we will condone the absence of the bases a even in the final line.

A1

4

[10]

Q12.

(a) $3 \ln x = 4$

$$\left(\ln x = \frac{4}{3} \right)$$

$$x = e^{\frac{4}{3}}$$

ISW. Condone $\sqrt[3]{e^4}$

B1

1

(b) $3 \ln x + \frac{20}{\ln x} = 19$

$$3(\ln x)^2 + 20 = 19 \ln x$$

correctly multiplying by $\ln x$.

M1

$$3(\ln x)^2 - 19 \ln x + 20 (= 0)$$

A1

$$(3 \ln x \pm 4)(\ln x \pm 5) (= 0)$$

use of formula, or completing the square must be correct

m1

$$\ln x = \frac{4}{3}, 5$$

A1

$$x = e^{\frac{4}{3}}, e^5$$

condone $\sqrt[3]{e^4}$

A1

5

[6]

Q13.

- (a) Use of one law of logs or exponentials

M1

$$\lg a = c \text{ and } \lg b = m$$

OE; both needed

A1

$$\text{So } a = 10^c \text{ and } b = 10^m$$

A1

3

- (b) Points (1, 1.08), (5, 1.43) plotted
M1 A0 if one point correct

M1A1

Straight line drawn through points
ft small inaccuracy

A1F

3

- (c) (i) Attempt at antilog of $Y(3)$
OE

M1

$$\text{When } x = 3, Y \approx 1.25 \text{ so } y \approx 18$$

Allow AWRT 18

A1

2

- (ii) Attempt at a as antilog of Y -intercept
OE

M1

$$a \approx 9.3 \text{ to } 10$$

AWRT

A1

2

[10]

Q14.

(a) (i) $(x =) 1$

CAO

B1

1

(ii) $(x =) 3$

CAO

B1

1

(b) $\log_a n^2 = \log_a 18(n - 4)$

A valid law of logs applied to correct logs

M1

A second valid law of logs applied to correct logs

M1

$n^2 - 18n + 72 = 0$

ACF of these terms eg $n^2 - 18n = -72$

A1

$(n - 6)(n - 12) = 0$

Valid method to solve quadratic, dep on both the previous Ms

m1

$n = 6, n = 12$

Both values required

SC NMS max (out of 5) B3 for both 6 and 12 without uniqueness considered; max B1 for either 6 or 12 only

A1

5

[7]

Q15.

(a) $(y =) 1$

B1

1

(b) $h = 0.2$

PI

B1

$$f(x) = 2^{4x}$$

$$I \approx h/2 \{...\}$$

$$\{.\} = f(0) + f(1) + 2[f(0.2) + f(0.4) + f(0.6) + f(0.8)]$$

OE summing of areas of the 'trapezia'.

M1

$$\{.\} = 1 + 16 + 2(2^{0.8} + 2^{1.6} + 2^{2.4} + 2^{3.2})$$

$$= 1 + 16 + 2(1.741.. + 3.031.. + 5.278.. + 9.1895..)$$

$$= [17 + 2 \times 19.24...]$$

OE Accept 2dp rounded or truncated evidence

A1

$$I = 5.55 \text{ (to 2 dp)}$$

Must be 5.55

A1

4

(c) Stretch(I) in y -direction(II) scale factor $\frac{1}{8}$ (III)

Need (I) and either (II) or (III)

M1

Need (I) and (II) and (III)

A1

ALTn: Translation with an indication
that the translation is in the x -direction (B1)

$$\begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}$$

(B1)

*Combination of **different**
transformations scores 0/2*

2

(d) $g(x) = 2^{4(x-1)} - \frac{1}{2}$

B1 for either $2^{4(x+1)} - \frac{1}{2}$ or for

$$2^{4(x-1)} + \frac{1}{2} \text{ or for } 2^{4x-1} - \frac{1}{2}$$

B2,1,0

At $Q, y = 0 \Rightarrow 2^{4(x-1)} = 2^{-1}$

Reaches a stage from which linear eqn can be stated directly eg an alternative stage is

$$4(x-1) \log 2 = -\log 2$$

M1

$$\Rightarrow 4x - 4 = -1 \Rightarrow x = 0.75$$

NMS mark as 4 or 0

A1

4

(e) (i) $\log_a k = \log_a 2^3 + \log_a 5 - \log_a 4$
 $\log_a k = \log_a (2^3 \times 5) - \log_a 4$

One law of logs used

M1

A second law of logs used; could be

$$\log_a k = \log_a 2^3 + \log_a \left(\frac{5}{4}\right)$$

M1

$$\log_a k = \log_a \left(\frac{2^3 \times 5}{4}\right) = \log_a 10 \Rightarrow k = 10$$

CSO AG

A1

3

(ii) $2^{4x-3} = \frac{5}{4}$ so

*Equate y's, take logs (to any base) of both sides **and** apply 3rd law of logs.*

$$(4x-3) \log_{10} 2 = \log_{10} \frac{5}{4}$$

M1

$$x = \frac{3 \log_{10} 2 + \log_{10} \left(\frac{5}{4} \right)}{4 \log_{10} 2}$$

m1

$$x = \frac{\log_{10} 10}{4 \log_{10} 2} \text{ so } x = \frac{1}{4 \log_{10} 2}$$

A1

Altn $4x \log 2 = \log \left(\frac{5}{4} \times 2^3 \right)$

Rearrange correctly to $x = \dots$

Altn $4x \log 2 = \log 10$

In both cases, log term(s) must have same base and expressions must be in an exact form, ie not approx. dec. vals

CSO AG Must be clear evidence that base 10 is used, also be convinced

3

[17]

Q16.

(a) (i) $\log_a 40$

Accept 'k = 40'

B1

1

(ii) $\log_a 8$

Accept 'k = 8'

B1

1

(iii) $\log_a 125$

Accept 'k = 125' but not 'k = 5³'

B1

1

(b) $\log_{10} \{(1.5)^{3x}\} = \log_{10} 7.5$

*Correct statement having taken logs of both sides of $(1.5)^{3x} = 7.5$ OE PI
or $3x = \log_{1.5} 7.5$ seen*

M1

$$3x \log_{10} 1.5 = \log_{10} 7.5$$

$$\log 1.5^{3x} = 3x \log 1.5 \text{ OE}$$

m1

$$x = \frac{\lg 7.5}{3 \lg 1.5} = 1.65645 \dots = 1.656 \text{ to 3 dp}$$

Both method marks must have been awarded with clear use of logarithms seen

A1

3

(c) $\log_2 p = m \Rightarrow p = 2^m; \log_8 q = n \Rightarrow q = 8^n$

Either $p = 2^m$ or $q = 8^n$ seen or used

M1

$$p = 2^m \text{ and } q = 2^{3n}$$

Writing $8^n = 2^{3n}$ and having $p = 2^m$

m1

$$pq = 2^m \times (2^3)^n = 2^m \times 2^{3n} \text{ so } pq = 2^{m+3n}$$

Accept $y = m + 3n$

A1

3

[9]

Q17.

(a) (i) $\sqrt{125} = \sqrt{25 \times 5} = 5\sqrt{5}$

OE eg $\sqrt{125} = \sqrt{5^3}$ or $5^{1.5}$ seen

M1

$$5^p = \sqrt{125} \Rightarrow p = 1.5$$

Correct value of p must be explicitly stated

A1

Alternative:

$$p \log 5 = \frac{1}{2} \log 125$$

2

OE eg $p \log 5 = \log 11.18$

or eg $p = \log_5 \sqrt{125}$

(M1)

$$p \log 5 = \frac{3}{2} \log 5 \Rightarrow p = \frac{3}{2}$$

Correct value of p must be explicitly stated

(A1)

(ii) $5^{2x} = \sqrt{125} = 5^{\frac{3}{2}} \Rightarrow x = 0.5p = 0.75$

Must be $0.5 \times c$'s value of p

SC: $x = 0.75$ with c 's ans (a)(i) $5^{1.5}$ scores B1F

B1F

1

(b) $3^{2x-1} = 0.05$

$(2x - 1) \log 3 = \log 0.05$

Take logs of both sides and use 3rd law of logs. PI eg by $2x - 1 = \log_3 0.05$ seen

M1

$$x = \frac{\log_{10} 0.05}{2 \log_{10} 3} + \frac{1}{2}$$

Correct rearrangement to $x = \dots$ PI

m1

$= -0.8634(165\dots) = -0.8634$ to 4dp

Condone > 4 dp. Must see logs clearly **used** in solution, so NMS scores 0/3

A1

3

(c) $\log_a x = 2(\log_a 3 + \log_a 2) - 1$
 $= 2 \log_a (3 \times 2) - 1$

A valid law of logs used

M1

$= \log_a (6^2) - 1$

Another valid law of logs used

M1

$= \log_a 36 - \log_a a$

$\log_a a = 1$ quoted or used

or $\log_a \frac{x}{k} = -1 \Rightarrow \frac{x}{k} = a^{-1}$ OE

B1

$$\log_a x = \log_a \left(\frac{36}{a} \right) \Rightarrow x = \frac{36}{a}$$

CSO Must be $x = \frac{36}{a}$ or $x = 36a^{-1}$

A1

4

[10]

Q18.

(a) $x = 8$

No clear log law errors seen.

Condone answer left as $\frac{16}{2}$

B1

1

(b) $\log_a y = \log_a 3^2 + \log_a 4 + 1$

One law of logs used correctly

M1

$$\log_a y = \log_a (3^2 \times 4) + 1$$

Either a second law of logs used correctly
or the 1 written as $\log_a a$

M1

$$\log_a y = \log_a (3^2 \times 4) + \log_a a = \log_a 36a$$

$$\Rightarrow y = 36a$$

CSO

A1

3

[4]

Q19.

(a) (i) $y = 2x^2 - 8x + 3$

$$\left(\frac{dy}{dx}\right)4x - 8$$

B1

1

$$\begin{aligned}
 \text{(ii)} \quad & \int_4^6 \frac{x-2}{2x^2-8x+3} dx \\
 &= \frac{1}{4} [\ln |2x^2-8x+3|]_4^6
 \end{aligned}$$

M1 for $k \ln (2x^2 - 8x + 3)$; allow $k \ln u$

M1A1

$$= \frac{1}{4} [\ln 27 - \ln 3]$$

*Correct substitution into
 $k \ln (2x^2 - 8x + 3)$ or 3, 27 into $k \ln u$*

m1

$$\begin{aligned}
 &= \frac{1}{4} \ln 9 \\
 &= \frac{1}{2} \ln 3
 \end{aligned}$$

A1

4

$$\begin{aligned}
 \text{(b)} \quad & \int x\sqrt{3x-1} dx \\
 & u = 3x - 1 \quad du = 3 dx \\
 & \text{OE}
 \end{aligned}$$

B1

$$\int = \left(\frac{1}{9}\right) \int \left(u^{\frac{3}{2}} + u^{\frac{1}{2}}\right) (du)$$

[2 terms in with rational indices

M1

$$= \left(\frac{1}{9}\right) \left[\frac{u^{\frac{5}{2}}}{\frac{5}{2}} + \frac{u^{\frac{3}{2}}}{\frac{3}{2}} (+c) \right]$$

Must be 2 terms with correct indices

$$\left(\text{only fit for } x = \frac{u-1}{3} \right)$$

A1F

$$= \frac{2}{45}(3x-1)^{\frac{5}{2}} + \frac{2}{27}(3x-1)^{\frac{3}{2}} + c$$

CSO OE

A1

4

[9]

Q20.

(a) $3 = k(3 + x + 3 - x)$

$$\text{OE } \frac{A}{3-x} + \frac{B}{3+x} \Rightarrow 6A = 3 \quad 6B = 3$$

M1

$$k = \frac{1}{2}$$

or eg put $x = 0, \frac{3}{9} = k\left(\frac{1}{3} + \frac{1}{3}\right) \Rightarrow k = \frac{1}{2}$

A1

2

(b) $\int_1^2 \frac{3}{9-x^2} dx = -\frac{1}{2} \ln(3-x) + \frac{1}{2} \ln(3+x)$

$$a \ln(3 \pm x)$$

M1

ft on k

A1F

$$= \frac{1}{2}((\ln 5 - \ln 1) - (\ln 4 - \ln 2)) = \frac{1}{2} \ln\left(\frac{5}{2}\right)$$

$$\text{accept } \ln\left(\frac{10}{4}\right)$$

$$\text{ft only for sign error in integral: } \frac{1}{2} \ln\left(\frac{5}{8}\right)$$

A1F

3

[5]

Q21.

(a) (i) $\log_a 1 = 0$

B1

1

(ii) $\log_a a = 1$

B1

1

(b) $\log_a x = \log_a (5 \times 6) - \log_a 1.5$

One law of logs used correctly

M1

$$\log_a x = \log_a \left(\frac{5 \times 6}{1.5} \right)$$

A second law of logs used correctly

M1

$$\log_a x = \log_a 20 \Rightarrow x = 20$$

A1

3

[5]

Q22.

(a) $3 \log_a x = \log_a 8 \Rightarrow \log_a x^3 = \log_a 8$

OE use of the log law

M1

$$x^3 = 8 \Rightarrow x = 2$$

A1

2

(b) $3 \log_a 6 - \log_a 8 = \log_a 6^3 - \log_a 8$

Correct use of one log law

M1

$$= \log_a \frac{6^3}{8}$$

Correct use of a different log law

M1

$$= \log_a \frac{216}{8} = \log_a 27$$

CSO AG (be convinced)

A1

3

(c) (i) $\{p = \} 3\log_{10} 3 - \log_{10} 8$

Substitute $x = 3$

M1

$$p = \log_{10} \frac{3^3}{8} = \log_{10} \frac{27}{8}$$

AG (be convinced)

A1

2

(ii) Gradient of $PQ = \frac{q-p}{6-3}$

$\frac{\text{difference in } y\text{-coords}}{\text{difference in } x\text{-coords}}$
used

M1

$$\dots\dots\dots = \frac{\log_{10} 27 - \log_{10} \frac{27}{8}}{3}$$

Any correct exact form

A1

$$\dots\dots\dots = \frac{1}{3} \log_{10} \left(27 \div \frac{27}{8} \right)$$

Correct use of log law

m1

$$\text{Gradient} = \frac{1}{3} \log_{10} 8 = \log_{10} 2$$

AG (be convinced)

A1

4

[11]

Q23.

(a) $\lg y = \lg a + b \lg x$

M1 for use of one log law

M1A1

2

(b) Use of above result

M1

$a = 10$

A1

$b = \text{gradient}$

OE; PI by answer $\pm \frac{1}{2}$

m1

$\dots = -\frac{1}{2}$

A1

4

[6]

Q24.

(a) $\log_a n = \log_a 3(2n - 1)$

OE Log law used PI by next line

M1

$\Rightarrow n = 3(2n - 1)$

*OE, but must **not** have any logs.*

m1

$\Rightarrow 3 = 5n \Rightarrow n = \frac{3}{5}$

A1

3

(b) (i) $\log_a x = 3 \Rightarrow x = a^3$

B1

1

- (ii) $\log_a y - \log_a 2^3 = 4$
 $3 \log 2 = \log 2^3$ seen or used any time in (ii)

M1

$$\log_a \frac{y}{2^3} = 4 \left\{ \begin{array}{l} xy = a^7 \times a^{(3 \log_a 2)} \\ \text{or} \\ y = a^4 \times a^{(3 \log_a 2)} \end{array} \right.$$

Correct method leading to an equation involving y (or xy) and a log but **not** involving + or -

M1

$$\frac{y}{2^3} = a^4 \left\{ \begin{array}{l} xy = a^7 \times 2^3 \\ \text{or} \\ y = a^4 \times 2^3 \end{array} \right.$$

Correct method to eliminate **ALL** logs e.g. using $\log_a N = k \Rightarrow N = a^k$ or using $a^{\log_a c} = c$

ml

$$by = a^3 \times 8a^4 \text{ or } 8a^7$$

A1

4

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[8]

Q25.

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- (a) $2 \log_a n - \log_a (5n - 24) = \log_a 4$
 $\Rightarrow \log_a n^2 - \log_a (5n - 24) = \log_a 4$
A law of logs used

M1

$$\Rightarrow \log_a \left[\frac{n^2}{5n - 24} \right] = \log_a 4$$

A second law of logs used leading to both sides being single log terms or single log term on LHS with RHS = 0

M1

$$\Rightarrow \frac{n^2}{5n-24} = 4$$

$$\Rightarrow n^2 - 20n + 96 = 0$$

CSO. AG

A1

3

(b) $\Rightarrow (n-8)(n-12) = 0$

Accept alternatives eg formula, completing of sq..

M1

$$\Rightarrow n = 8, 12$$

A1

2

[5]

Q26.

(a) $\log_a x = \log_a 6^2 - \log_a 3$

One law of logs used correctly

M1

$$\log_a x = \log_a \left(\frac{6^2}{3} \right)$$

A second law of logs used correctly

M1

$$\log_a x = \log_a \frac{36}{3} \Rightarrow x = 12$$

CSO AG

A1

3

(b) $\log_a y + \log_a 5 = 7 \Rightarrow \log_a 5y = 7$

M1

$$\Rightarrow 5y = a^7 \text{ or } y = \frac{1}{5} a^7 \text{ or } a = (5y)^{1/7}$$

Eliminates logs

m1

Accept these forms

A1

3

[6]

Q27.

(a) $\log_a x = \log_a 6^3 - \log_a 8$

A law of logs used correctly

M1

$$\log_a x = \log_a (6^3 \div 8)$$

A different law of logs used correctly

M1

$$x = 6^3 \div 8 = 27$$

CSO AG (be convinced)

A1

3

ALT $\log_a x = 3 \log_a 6 - 3 \log_a 2$

(M1)

$$\frac{1}{3} \log_a x = \log_a \frac{6}{2}$$

(M1)

$$x^{\frac{1}{3}} = 3 \Rightarrow x = 27$$

©2025 Exam Papers Practice. All rights reserved. (A1) CSO

(b) (i) $\log_4 1 = 0$

B1

(ii) $\log_4 4 = 1$

SC in (b): For all four answers $\frac{1}{4}$; 1; $\frac{1}{2}$; 2

B1

(iii) $\log_4 2 = 0.5$

give 0/4; otherwise mark each

B1

(iv) $\log_4 8 = 1.5$

independently

B1

4

[7]

Q28.

(a) $\frac{a}{1-r} = 4a$

(Accept $S_{\infty} = \frac{\frac{a}{3}}{1-\frac{3}{4}}$)

M1

$\Rightarrow 1-r = \frac{a}{4a}$ or $a = 4a(1-r)$
 Either (or better) (or $S_{\infty} = 4a$ if M1 above)

A1

$1-r = \frac{1}{4} \Rightarrow r = \frac{3}{4}$

AG Be convinced. (or statement 4 times 1st term)

A1CSO

(b) $(S_{10} =) \frac{48(1-r^{10})}{1-r}$

Correct formula with $n = 10$ and one of

$a = 48$ or $r = \frac{3}{4}$ OE

M1

$= 192(1 - 0.75^{10}) = 181.1878$ to 4dp

A1

2

(c) (i) $u_n = ar^{n-1} = a \left(\frac{3}{4}\right)^{n-1} = 48 \left(\frac{3}{4}\right)^{n-1}$

B1

$u_{2n} = ar^{2n-1} = a \left(\frac{3}{4}\right)^{2n-1} = 48 \left(\frac{3}{4}\right)^{2n-1}$
 ft on candidate's $u_n = ar$ function of n

B1ft

2

$$(ii) \quad \frac{u_n}{u_{2n}} = \frac{ar^{n-1}}{ar^{2n-1}} = \frac{r^{n-1}}{r^{2n-1}}$$

Eliminating a (or 48) or log a

M1

$$\log_{10} u_n - \log_{10} u_{2n} = \log_{10} \frac{u_n}{u_{2n}}$$

Using at least one log law

M1

$$= \log_{10} \frac{r^{n-1}}{r^{2n-1}} = \log_{10} (r^{-n})$$

$$= -n \log_{10} \frac{3}{4} = n \log_{10} \frac{4}{3}$$

A1CSO

AG Full valid completion

3

$$(iii) \quad \log_{10} \left[\frac{u_{100}}{u_{200}} \right] = 100 \log_{10} \left(\frac{4}{3} \right)$$

M1

$$= 12.49\dots = 12.5 \text{ to 3 sf}$$

AG Be convinced

SC: Those applying 'hence' to (i) rather than to (ii)

Mark as B2

A1CSO

2

[12]

Q29.

$$(a) \quad (1+x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$$

Full method

M1

A1 if four terms correct or just one slip

A2,1

3

(b) (i) $(1 + \sqrt{5})^4 = 1 + 4\sqrt{5} + 6(\sqrt{5})^2 + 4(\sqrt{5})^3 + (\sqrt{5})^4$
Substitute. $\sqrt{5}$ for x .

M1

$$= 1 + 4\sqrt{5} + 6(5) + 4(5\sqrt{5}) + (25)$$

Two of 3 terms shown in brackets

A1ft

$$= 56 + 24\sqrt{5}$$

AG *Be convinced*

A1CSO

3

(ii) $\log_2 (1 + \sqrt{5})^4 = \log_2 [8(7 + 3\sqrt{5})]$

M1

$$= \log_2 8 + \log_2 (7 + 3\sqrt{5})$$

m1

$$= 3 + \log_2 (7 + 3\sqrt{5})$$

SC B1 *Change to base 10 and verify as per model marked script*

A1CSO

3

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[9]

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